

Assignment 1

LLM Architecture Analysis

Course Code: CSAI601	Assessment: Assignment 1/5	Total Marks: 4 Marks	CO Mapped: CO1
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Objective: Demonstrate understanding of transformer architecture and Constitutional AI.

Submission: PDF report (max 6 pages) + Python script (.py file) via LMS. **Due:** End of Week 2.

Q1. [1 Mark] Draw a detailed diagram of the Transformer Encoder block and label: multi-head attention, feed-forward network, residual connections, and layer normalisation. Explain the role of each component in 3–4 lines each.

Q2. [1 Mark] Explain the self-attention mechanism mathematically. Given Query (Q), Key (K), and Value (V) matrices, derive the attention output. Why is the scaling factor $1/\sqrt{d_k}$ necessary? What happens without it?

Q3. [1 Mark] Compare RLHF and Constitutional AI (RLAIF) using a structured table with the following columns: (a) Training pipeline steps, (b) Data requirements, (c) Scalability, (d) Key limitation. After the table, write a 100-word paragraph arguing which approach is better and why.

Q4. [1 Mark] Write a Python script that: (i) authenticates with the Anthropic API using an environment variable, (ii) sends a 5-message conversation history to Claude, (iii) prints the text response, (iv) prints input and output token counts, (v) handles rate limit errors with a 10-second retry. Submit the script and a screenshot of successful output.

Evaluation Rubric: Accuracy of diagram/explanation (35%), Mathematical correctness (25%), Depth of comparison (25%), Code functionality (15%)

Assignment 2

Prompt Engineering Portfolio

Course Code: CSAI601	Assessment: Assignment 2/5	Total Marks: 4 Marks	CO Mapped: CO2
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Objective: Apply and compare different prompting strategies on real tasks.

Submission: Jupyter Notebook (.ipynb) + PDF report. **Due:** End of Week 4.

Q1. [1 Mark] Choose any text classification task (sentiment analysis, topic classification, spam detection). Implement three versions: (a) zero-shot, (b) 3-shot, (c) 5-shot. Test each on 20 labelled samples. Report accuracy in a table. Analyse: does adding more examples always improve performance?

Q2. [1 Mark] Design a Chain-of-Thought prompt for the following problem: "A train travels 120 km in 2 hours, stops for 30 minutes, then travels 80 km at 60 km/h. What is the total journey time?" Show: (a) the complete prompt, (b) Claude's full CoT response, (c) apply Self-Consistency (3 independent runs) and report the majority answer.

Q3. [1 Mark] Write a system prompt that creates a "Senior Python Code Reviewer" persona. The persona should: identify bugs, suggest PEP 8 improvements, and rate code quality 1–10. Test on 3 code snippets of varying quality. Create a rubric to evaluate the persona's output quality.

Q4. [1 Mark] Implement the ReAct framework manually (no agent loop code). Write a prompt that instructs Claude to interleave "Thought: ...", "Action: ...", "Observation: ..." steps to answer: "What is the population of the capital city of the country that won the 2022 FIFA World Cup?" Provide 3 fully worked examples.

Note: Use claude-3-haiku-20240307 to manage costs. Include token counts for each experiment.

Assignment 3

Build a Claude-Powered Application

Course Code: CSAI601	Assessment: Assignment 3/5	Total Marks: 4 Marks	CO Mapped: CO3
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Objective: Develop a working application using the Claude API. Individual submission.

Submission: GitHub repo link + 2-page write-up. **Due:** End of Week 6.

Choose ONE track:

Q1. [4 Marks — Track A: RAG Document Assistant] Build a system that: (i) ingests a PDF (min 20 pages), (ii) chunks it using a strategy you choose and justify, (iii) embeds chunks using any embedding model, (iv) stores vectors in ChromaDB or FAISS, (v) answers user questions using Claude with the top-3 retrieved chunks injected. Evaluate on 10 questions with and without RAG — measure answer quality.

Q1. [4 Marks — Track B: Tool-Use Agent] Build a Claude agent with at least 3 tools (e.g., web search, calculator, weather API, Wikipedia). Implement the tool loop manually using the Anthropic SDK (no LangChain). Demonstrate 5 multi-step queries where Claude uses multiple tools in sequence.

Q1. [4 Marks — Track C: Vision Application] Build an app that accepts image input and answers questions. Implement batch processing for 10 images. Add a Streamlit or Gradio web UI. Demonstrate at least 3 use cases (e.g., chart reading, document OCR, scene description).

All tracks must include: (a) Clean Python code with docstrings, (b) README with setup and run instructions, (c) Cost estimation for 100 API calls, (d) Error handling for at least 3 failure modes.

Assignment 4

AI Ethics Case Study Report

Course Code: CSAI601	Assessment: Assignment 4/5	Total Marks: 4 Marks	CO Mapped: CO4
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Objective: Critically analyse a real-world AI ethics incident.

Submission: PDF report, max 8 pages. **Due:** End of Week 8.

Q1. [1 Mark] Select a documented AI ethics incident (not discussed in class). Write a 500-word case description covering: what happened, who was affected, what the technical root cause was, and what the company's response was. Include references.

Q2. [1 Mark] Analyse the incident using TWO frameworks: (a) Anthropic's HHH (Helpful, Harmless, Honest) — rate the system on all three dimensions, and (b) EU AI Act risk categories — classify the system and justify. Does this system qualify as "high risk"? Why or why not?

Q3. [1 Mark] Identify 3 specific technical changes that could have prevented the incident. For each change: describe the implementation, explain the trade-off it introduces, and rate its feasibility (Easy/Medium/Hard).

Q4. [1 Mark] Design a Responsible AI Checklist (minimum 12 items, max 20) for a developer building a healthcare AI assistant powered by Claude. Organise by category: Data Governance, Model Behaviour, Deployment, Ongoing Monitoring. Each item must be actionable (yes/no checkable).

Assignment 5

Capstone Project Proposal & Architecture

Course Code: CSAI601	Assessment: Assignment 5/5	Total Marks: 4 Marks	CO Mapped: CO5
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Objective: Plan the Capstone Project to be demonstrated in Week 15. Groups of 2–3 students.

Submission: PDF document + architecture diagram image. **Due:** End of Week 10.

Q1. [1 Mark] Problem Statement: (a) Identify a real-world problem solvable with Claude. (b) Justify why an LLM is the appropriate solution vs a rule-based system, traditional ML, or manual approach. (c) Define 3 measurable success criteria (e.g., "answers 80% of test questions correctly").

Q2. [1 Mark] System Architecture: Draw a complete architecture diagram showing: User Interface, Claude API, all tools/databases/external APIs, caching layer, deployment environment. Label all data flows with arrows. Describe each component in 2–3 lines.

Q3. [1 Mark] Prompt Design: Write the complete system prompt for your application. Write 2 example user messages with Claude's expected ideal response. Justify your prompting strategy (which technique and why).

Q4. [1 Mark] Ethics & Risk Analysis: (a) Identify 3 ethical risks of your application. For each: describe the risk, rate severity (Low/Medium/High), and specify a technical mitigation. (b) Calculate estimated Claude API cost for 1,000 daily active users, each sending 5 queries of average 500 input tokens and 200 output tokens. State which model you'll use and why.